Copernicus Global Land Operations - Lot I

"Vegetation and Energy"

Framework Service Contract N° 945120 – IPR – 2023 (JRC)

ALGORITHM THEORETICAL BASIS DOCUMENT

10-DAY LAND SURFACE TEMPERATURE – LST10

VERSION 2.2

ISSUE I1.91

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11.09.2025	10	Update to LST10 V2.2	I1.91

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LIST OF ACRONYMS

AD	Applicable Document
ATBD	Algorithm Theoretical Basis Document
CAL/VAL	Calibration and Validation
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
ECMWF	European Centre for Medium-Range Weather Forecasts
ERA	ECMWF Re-Analysis
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
GCOS	Global Climate Observing System
GEO	Geostationary satellites
GEWEX	Global Energy and Water cycle Experiment
GOES	Geostationary Operational Environmental Satellite
INSPIRE	Infrastructure for Spatial Information in Europe
IODC	Indian Ocean Data Coverage
IPMA	Instituto Portugês do Mar e da Atmosfera
ITCZ	Inter-Tropical Convergence Zone
JRC	Joint Research Center
LST	Land Surface Temperature
MIR	Middle infrared
MSG	Meteosat Second Generation
MTSAT	Multi-Function Transport Satellite
NDVI	Normalized Difference Vegetation Index
NWC SAF	EUMETSAT SAF on Nowcasting and Very Short Range Forecasting
PUM	Product User Manual
QAR	Quality Assessment Report
SAF	Satellite Application Facility
SEVIRI	Spinning Enhanced Visible Infra-Red Imager
SKT	Skin Temperature
SSD	Service Specification Document
STD	Standard Deviation
SVP	Service Validation Plan
TCC	Total Cloud Cover
TCI	Thermal Condition Index
TIR	Thermal Infrared
UML	Unified Modelling Language
UTC	Coordinated Universal Time
WGS	World Geodetic System
WMO	World Meteorological Organisation

EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery take place in a timely manner and are complemented by the constitution of long-term time series.

This document describes the procedures used to calculate a 10-day synthesis of global Land Surface Temperature (LST) obtained from a constellation of geostationary (GEO) satellite missions: Meteosat Second Generation (MSG) 0º and Indian Ocean Data Coverage (IODC); Geostationary Operational Environmental Satellite (GOES) and Himawari (or its predecessor, the Multi-Function Transport Satellite MTSAT) benefiting from high temporal frequency and wide area coverage at hourly rate at 5/112° resolution.

1. BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The aim of 10-day LST product is to provide a complete overview of the LST daily cycle over each 10-day compositing for every image pixel. The composites contain the maximum, median and minimum LST values observed per time slot (i.e., 00, 01, …, 23 UTC) during the 10-day period. Those composites can be used to retrieve a synthesis of LST conditions and to derive a Thermal Condition Index (TCI), an index which characterizes the pixel temperature within its expected maximum range. Therefore a 10-day TCI is also provided by the 10-day LST product to ease the use of such information in e.g. agrometeorological applications.

The TCI calculation requires a climatology of LST values around local noon per pixel, which was specifically estimated for this purpose using all processed and archived LST fields and the ECMWF ERA-Interim reanalysis. The methodology used to obtain this long-term series is also presented in this document.

The 10-day LST accuracy is, at least, that of the nominal hourly product, i.e., root mean square differences with respect to 10-day composites of in situ temperature should be within 4 K. The quality assessment of the 10-day LST composites is used as an indication of the TCI validation.

This document describes the theoretical basis of the 10-day LST V2.2 retrieval methodology. Note that there is no change in the algorithm between LST/LST10 V2.2 and LST/LST10 V2.1; only some attributes of the metadata have been fixed (see Table 5 in [CGLOPS1_ATBD_LST-V2.2] for details).

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Section 2 recalls the user requirements applicable for LST10 product.
- Section 3 gives a brief overview of the product and the strategy used on the product retrieval.
- Section 4 describes the algorithms and the input data used to retrieve the LST10. This section also contains examples of variables contained in the output file product.
- Section 5 states the assumptions and main limitations of the LST10 product.
- Section 6 enumerates the risks and respective mitigation procedures.

1.3 RELATED DOCUMENTS

1.3.1 Applicable document

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_ATBD_LST-V2.2	Algorithm Theoretical Basis Document of Land Surface Temperature Product Version 2.2
CGLOPS1_PUM_LST-V2.2	Product User Manual for Land Surface Temperature Version 2.2

1.3.3 Output

Document ID	Descriptor
CGLOPS1_PUM_LST10-V2.2	Product User Manual for 10-day Land Surface Temperature Version 2.2
CGLOPS1_QAR_LST-LST10-V2.1	Quality Assessment report for Land Surface Temperature product family Version 2.1

1.3.4 External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs requirements.

Available online at https://library.wmo.int/

2. REQUIREMENTS

According to the applicable document [AD1], the specific technical details and requirements are described below (Table 1).

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

Table 1 – CGLOPS LST product specifications.

LST – LAND SURFACE TEMPERATURE	
The Land Surface Temperature (LST) is the radiative skin temperature of the land surface, as measured in the direction of the remote sensor [K].	
Geometric Properties:	
Spatial resolution	5 km
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetical datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage	Global
Time span	Hourly and 10-day period
Timeliness	Hourly: within 4 hours; within 2 days (optimally 1 day) after the end of 10-day period
Uncertainty (2-sigma)	Threshold: 2 K, Goal: 1 K
Stability (per decade)	Threshold: 0.3 K, Goal: 0.1 K

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.

Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product;
- The per-pixel date of the individual measurement or the start-end dates of the period covered;
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result;

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets shall be available in NETCDF. Additional format, such as COG or ZAR, format can be proposed for production or requested by the contracting authority during the Framework.

2.2.4 Uncertainties and validation

Uncertainties indicated in the product specifications (Table 1) follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate, the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL datasets and/or Ground-Based Observations for Validation (GBOV,

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, and conversion to physical values formulas.

https://land.copernicus.eu/en/products/GBOV) of Copernicus Global Land component when biophysical parameters are available.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu;

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary datasets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination. The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3. LST10 OVERVIEW

The Copernicus Global Land Service provides an LST retrieved from a constellation of geostationary (GEO) missions – Meteosat Second Generation (MSG) 0º and Indian Ocean Data Coverage (IODC); Geostationary Operational Environmental Satellite (GOES); and Himawari-8 (or its predecessor Multi-Function Transport Satellite – MTSAT) benefiting from high temporal frequency and wide area coverage at hourly rate at 5/112º resolution covering land pixels between 60°S and 70°N.

The latitude limits of 60°S and 70°N correspond to the limits of land regions that can be observed by the GEO satellites. Higher latitudes can only be observed by polar orbiting satellites. Thus, the global CLMS LST could benefit from the inclusion of LST derived from polar orbiter missions, such as Sentinel-3.

To ease integration of the LST products together with other CLMS variables in, e.g., agrometeorological applications, the Copernicus Global Land Service offers a 10-day composite of LST: LST10. This product consists of a complete overview of the LST daily cycle per pixel over each 10-day compositing period. Additionally, a 10-day synthesis and Thermal Condition Index (TCI) is also calculated and distributed with the LST10 product.

The hourly maps of global LST are used to estimate the main parameters characterizing the diurnal cycle of LST for each pixel – the maximum, the minimum, the median temperatures – as well as the fraction of valid pixels used in the calculation. These parameters are estimated for each of the 10 days. Statistics for the 10-day period are then derived and compared to a multi-year climatology to produce TCI.

4. ALGORITHM DESCRIPTION

4.1 PROCESSING OUTLINE

The LST10 product is generated every 10 days (at the 1st, 11th, 21th days of the month), with a timeliness up to 3 days, through the following steps:

Step 1: gather the LST product of all timeslots of the dekad - corresponding to 10 days, except for the last dekad of the month where the period ranges between 8 and 11 days depending on the number of days of the month.

Step 2: Remove positive and negative outliers from the clear sky LST observations keeping only valid LST values.

Step 3: Calculate the decadal statistical parameters (maximum, median, minimum) of the LST product per pixel.

Step 1 to Step 3 are repeated for all the possible timeslots (00, 01, …, 23UTC) to generate a set of 24 maps of statistical parameters obtained for the specific timeslot during the compositing period. The statistical parameters include the maximum, minimum and median for the whole dekad. In addition, a map with the fraction of valid observations is also provided.

Step 4: Calculate a synthesis of the LST product over the dekad, regardless the time of the day. The synthesis includes the maximum, median and minimum values of LST. The fraction of valid observations in the period is also provided as an ancillary dataset.

Step 5: Calculate the Thermal Condition Index (TCI) obtained by comparing the median of the LST values observed during the compositing period around the local solar noon with the upper and lower values of maximum LST from a multi-year climatology.

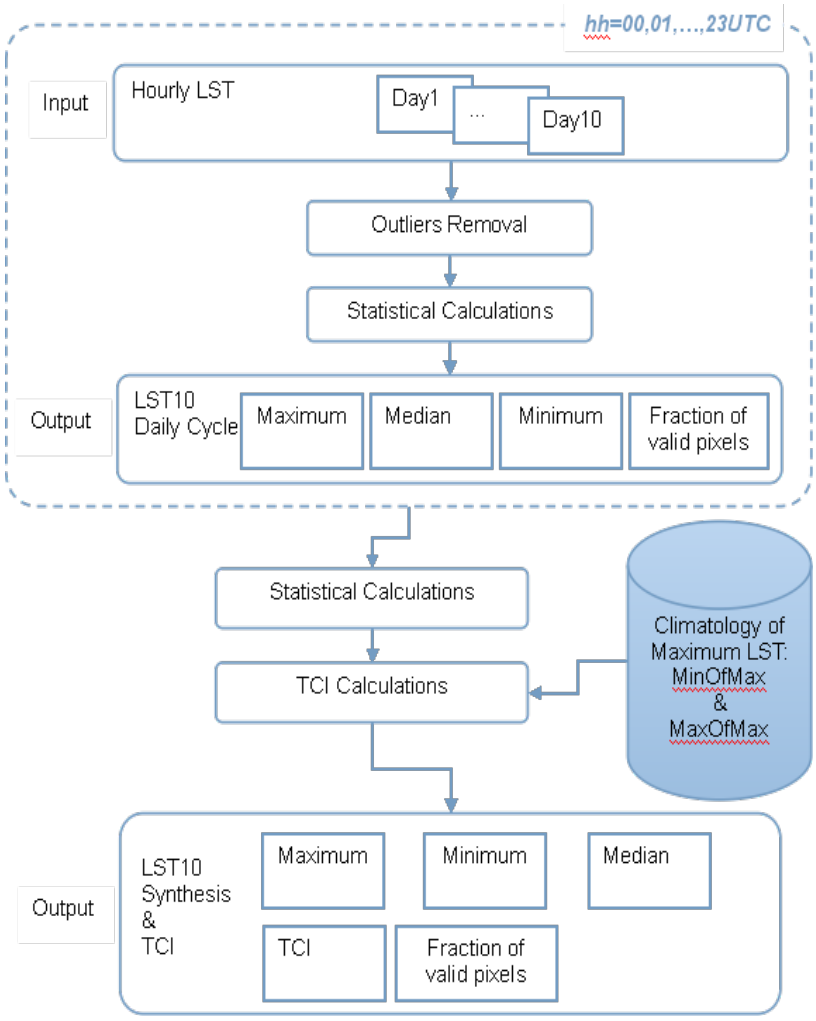


Figure 1 - Flow diagram of the generation of 10-day LST products.

4.2 INPUT DATA

4.2.1 LST

Land Surface Temperature (LST) is estimated from Top-of-Atmosphere (TOA) brightness temperatures of atmospheric window channels within the infrared range. The algorithms developed for GEO satellites take into account the information in the available channels within the thermal infrared range (T_{TIR1} and T_{TIR2}) and in the middle infra-red (T_{MIR}) – around 10.8, 12.0 and 3.9 µm, respectively. The latter is only used for night-time conditions, when it is not contaminated by solar radiation reflected by the surface (and only for older sensors such as GOES-13, where a second TIR channel was not available).

The LST retrieval methodology (described in the product ATBD [CGLOPS1_ATBD_LST-V2.2]) is based on semi-empirical formulations, where LST is expressed as a regression function of TOA brightness temperatures obtained in clear sky conditions. The cloud screening

is based on a multispectral threshold technique applied to visible and infrared channels available in GEO sensors. The set of tests applied to GOES and HIMAWARI (or MTSAT) data correspond to a subset of those used for the Spinning Enhanced Visible and Infrared Imager (SEVIRI/MSG) channels implemented in the Nowcasting and Very Short-Range Forecasting (NWC) SAF software, as described in Annex 3 of the LST product ATBD [CGLOPS1_ATBD_LST-V2.2]. From LST-V2.1 onwards, surface emissivity accounts for snow cover, which can affect LST and its diurnal cycle amplitude in particular. The impact of this change is documented in the Quality Assessment Report [CGLOPS1_QAR_LST-LST10-V2.1].

The main input to the LST10 is the hourly LST global product provided by Copernicus Global Land service in the projection and resolution described in the LST10 requirements (Chapter 2).

4.2.2 Climatology of maximum LST

Given the short time record of LST satellite retrievals within the Copernicus Global Land Service, extreme values used for TCI computation are based on a long-term series specifically built for this purpose. Those extreme values correspond to upper and lower values which would have occurred around the solar midday (i.e., around local daily LST maximum) for each pixel, during a 30-year period between 1985 and 2015.

The following steps describe the methodology used for building the series. The respective Unified Modelling Language (UML) Activity diagram is shown in Figure 2:

- Gather 30-year skin temperature (SKT) and total cloud cover (TCC) variables from ECMWF ERA-Interim reanalysis. The data are available at 0.75ºx0.75º spatial resolution and every 3 hours.
- Select the SKT and TCC data corresponding to the interval: [local solar noon – 1 hour, to local noon + 2 hours]. This is the same timeframe used for TCI estimation (section 4.5).
- Interpolate SKT to a 5/112º x 5º/112º regular grid. An atmospheric temperature lapse rate of -0.67ºC/100m is applied to correct for the difference between model orography and pixel height; cloudy pixels are excluded since LST is a clear-sky variable.
- Perform a linear regression analysis between LST and the corrected SKT ($SKT_{corrected}$) for the period that matches all the archived LST data (5 years). Linear regression coefficients for each pixel, a and b, are estimated, as well as the respective sample size (N) and the coefficient of determination, R^2, to evaluate the goodness of fit.

- Estimate virtual LST ($LST_{virtual}$), applying the linear regression coefficients to the *corrected SKT* data:

$$LST_{virtual} = a.SKT_{corrected} + b \quad (1)$$

Only robust regressions are used, depending on the combination of N and R^2 values per pixel. Based on the analysis of N and R^2 spatial distribution, the following criterion was established: N > 20 and R^2 > 0.7. The abovementioned criterion is a compromise between robustness of the regressions, on one hand, and availability of valid regressions that allow the estimation of "a climatological LST" from ERA-Interim. In a first step, we only considered regression fits when at least 30 pairs were available. However, we identified a number of areas where R^2 was still relatively large (above 80%) despite the very low number of available matchups. This suggested that the threshold for N could be relaxed. On the other hand, lower values of R^2 are found mostly in the tropics: a visual inspection of the spatial distribution of R^2 reveals that, as expected, low values are clustered in areas where cloud cover is more frequent. We apply the above linear regression to build a dataset of virtual LST on a pixel-by-pixel basis and for the full 1985-2015 period, for pixels where the criterion is satisfied. The remaining pixels are filled with the respective *corrected SKT* value. This includes the area now covered by the MSG IODC mission, since data from that mission is only available from June 2017 onwards. Thus, the 30-year climatology uses both measures, *virtual LST* and *corrected SKT*. The coverage of valid regressions is shown in section 5.1 (Figure 3).
Since the IODC LST is available for a very short period (< 5 years), the LST climatology used in this area is based on only ERA-Interim SKT; this climatology will be improved as soon as a sufficiently long LST database is available.

- The extreme values per dekad for each pixel are then the absolute maximum and absolute minimum between *virtual LST/corrected SKT* (30 years) and LST (5 years), around the respective solar noon (solar noon-1h to solar noon+2h).

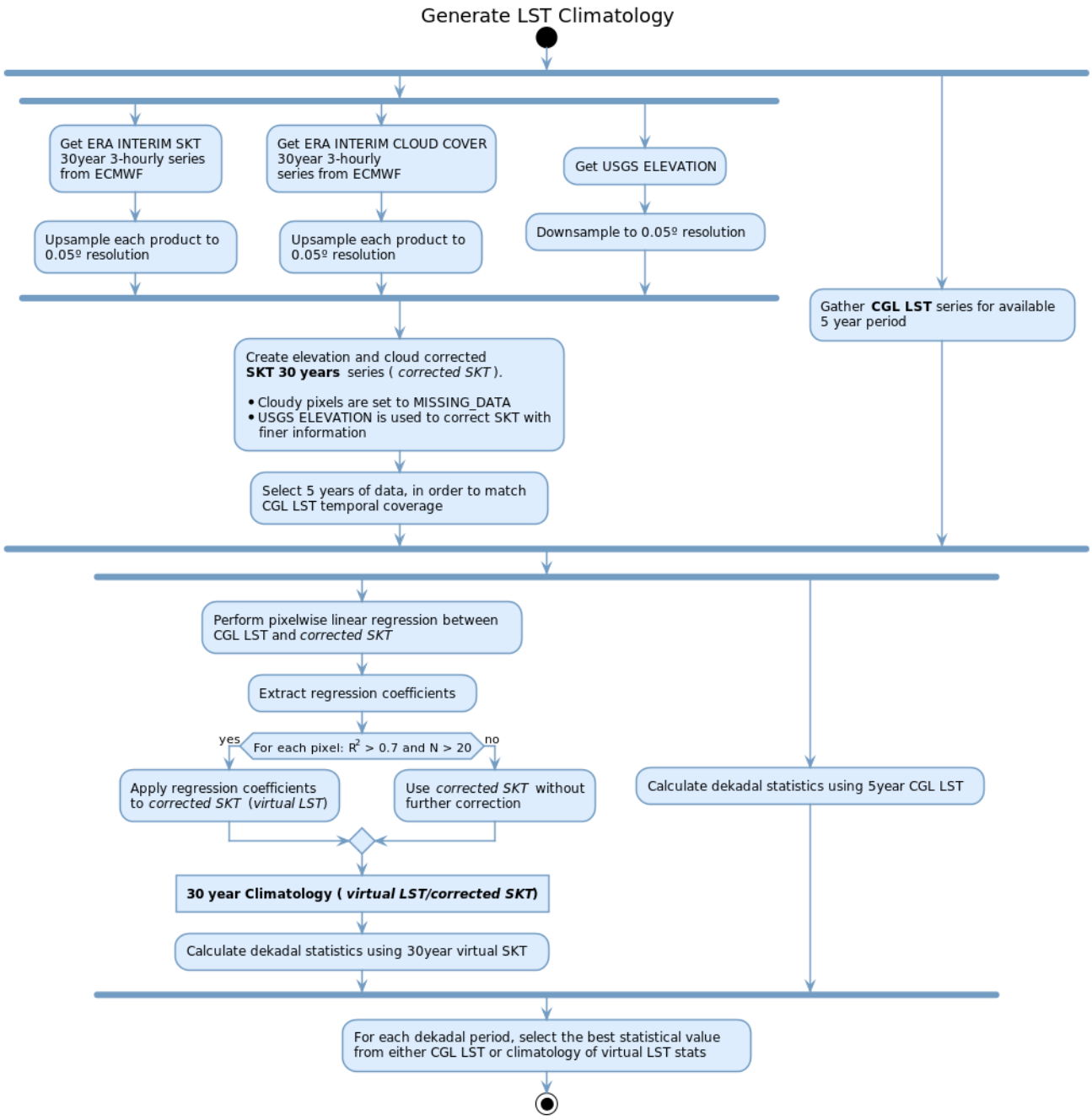


Figure 2 - UML Activity diagram describing the procedure for building the LST 30-year climatology and respective extremes values.

4.3 OUTLIER REMOVAL

The first step to remove LST outliers is to assume that valid values are within the physical range as defined in the LST PUM [CGLOPS1_PUM_LST-V2.2]: [-70ºC, 80ºC]. This procedure eliminates most of the unrealistic LST values caused by wrong values in the satellite

imagery (e.g. image stripes) but does not remove other outliers, e.g. LST retrievals obtained in pixels contaminated by clouds. Also does not remove other effects, such as fires or volcanoes, which may lead to LST retrievals within the physical range but several degrees higher than expected for the region and time of the day. This is particularly relevant for the retrieving of maximum LST value which, in such case would not be representative of the compositing period.

A spatio-temporal filter is applied to the LST product to remove both positive and negative outliers. The filter uses the LST pixel value and its neighbours (in a box of 3x3 pixels) over a dekad to calculate the mean and standard deviation (STD). The pixels outside a range of two times the STD around the mean value are disregarded since they are considered not being representative of dekad.

The outlier removal procedure described above is performed previous to LST10 product calculations described in the following sections (4.4 and 4.5).

Outliers of TCI variable, i. e. caused by temperatures outside the range of the 30-year climatology, cause the saturation of the index, i. e. they are set as 0 or 1, in case they are less than 0 or greater than 1, respectively. This is done from V2.0 onwards; for previous product versions these values were masked out.

4.4 LST10 – Daily Cycle

One of the main advantages of retrieving LST from GEO satellites is the ability to characterize the LST daily cycle. A temporal composite of LST is aimed for applications that do not require information on the daily variation of the LST within a relatively short period and/or need an LST daily cycle without gaps due to cloudiness.

The LST10 – Daily Cycle is a 10-day composite of LST product for hourly timeslots (00 to 23UTC) and provides maps of the following parameters:

- Maximum, median and minimum of valid LST values observed during the compositing period per timeslot. In case of very few valid values (< 5) those statistics are not calculated since they would not be representative of dekad and could lead to unreliable results.
- Fraction of valid LST values (used to calculate statistics) in the compositing period. In this context, a valid LST value corresponds to clear-sky retrieval, after outlier removal.

Despite the outlier removal, the product provides the median rather than the average of LST, since the former is less sensitive to outliers especially for smaller sample sizes.

4.5 LST10 – SYNTHESIS & TCI

During the last decades, satellite data has been used to monitor crops and pasture development (e.g., Kogan et al., 2012). It has been demonstrated that vegetation health indices may be derived from moisture and thermal conditions (e.g., Kogan, 2001). Therefore, the 10-day LST composites described above can be used to provide a synthesis of LST conditions of the compositing period and to derive 10-day Thermal Condition Index (TCI):

$$TCI = \frac{[LST_{Max}]_{Max} - LST_{10dayMax}}{[LST_{Max}]_{Max} - [LST_{Max}]_{Min}} \quad (2)$$

Where $LST_{10dayMax}$ is the median of the LST values around the local solar noon, over a dekad, and where $[LST_{Max}]_{Max}$ and $[LST_{Max}]_{Min}$ are the extremes of maximum LST as in the climatology of maximum LST (section 4.2.2). $LST_{10dayMax}$ corresponds to a smoothed LST maximum over each dekad, in line with the definition first proposed by Kogan (1997). Nevertheless, it is worth mentioning that this TCI product is an adaptation of the model proposed by Kogan (2001, 1997) to geostationary satellite observations.

TCI, together with vegetation parameters such as the Vegetation Condition Index derived from the Normalized Difference Vegetation Index (NDVI), has long been used as a suitable indicator of vegetation health. Following the equation above, TCI characterizes how warm any given land pixel is with respect to its maximum temperature range. Since over vegetated surfaces, temperature is strongly controlled by energy fluxes (sensible and latent heat), TCI indirectly characterizes the moisture availability through the near-surface radiation and aerodynamic conditions (Anderson et al., 2007; Kogan et al., 2012).

It should be noted that TCI is generally estimated from polar-orbiter satellites using the daytime LST or brightness temperatures of the daytime passage. The use of geostationary platforms and corresponding increase in temporal sampling is expected to smooth and increase the reliability of TCI values. The number of clear observations available during the compositing period provides further indication of the representativeness of the composite values.

Together with TCI and the 10-day LST synthesis datasets, ancillary datasets for complementary information are provided, including:

- Fraction of valid LST values (used to calculate statistics) in the compositing period. In this context, a valid LST value corresponds to clear-sky retrieval, after outlier removal.

- Maximum, median and minimum LST values observed during the compositing period (regardless of the observation time). In case of very few valid LST values (<5) those statistics are not calculated since they would not be representative of the dekad and could lead to unreliable results.

The $LST_{10DayMax}$ used in the TCI calculations corresponds to the median of the LST value observed during the compositing period in a timeframe between the local noon minus 1 hour and local noon plus 2 hours since the maximum LST tends to occur shortly after the peak of solar heating. The local noon is determined by the time at the solar zenith angle reaches its minimum value in a specific day. Unavailable LST within that timeframe, during the whole dekad implies missing TCI values.

4.6 ALGORITHM OUTPUT

The 10-day LST product output is composed by 2 files, 1 file holding a composite of the 10-day period for each hour of a day (Daily Cycle), plus 1 file aggregating every hour of the respective decade (Synthesis & TCI) as described below.

LST10 – Daily Cycle

Each file is composed by a set of variables for a specific timeslot (00, 01, …, 23 UTC):

- **MAX**, **MEDIAN** and **MIN**: respectively the maximum, median and minimum LST values observed, at a specific timeslot, during the compositing period;
- **FRAC_VALID_OBS**: Fraction of valid observations as used to calculate the statistical parameters, for each specific timeslot

LST10 – Synthesis & TCI

The file is composed by a set of variables for the compositing period:

- **MAX, MEDIAN** and **MIN**: respectively the maximum, median and minimum LST values observed during the compositing period, regardless the time of the day;
- **FRAC_VALID_OBS**: Fraction of valid observations as used to calculate the statistical parameters, regardless the time of the day;
- **TCI**: Thermal Condition Index for the compositing period, estimated using LST observations around local solar noon;

The fraction of valid observations used to calculate the LST composite (FRAC_VALID_OBS) can be considered as a quality indicator to decide whether a pixel value should be used or disregarded. Different thresholds might be applied depending on the product usage.

A full description of the product file format and content may be found in the LST10 PUM [CGLOPS1_PUM_LST-V2.2].

5. ASSUMPTIONS AND LIMITATIONS

5.1 ASSUMPTIONS

It is assumed that the maximum LST value for a specific day occurs around local noon – when the solar zenith angle reaches its minimum.

It is assumed that the *virtual LST* can be used to represent the LST climate for geographic regions satisfying the condition (R^2 >= 0.7 and N >= 20), as well as the *corrected SKT*, for the remaining regions.

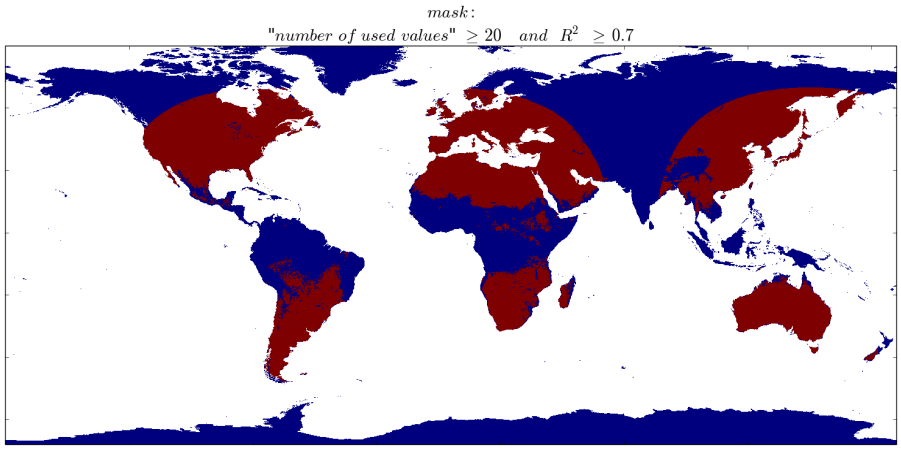


Figure 3: *Virtual LST* mask. Red colored regions satisfy the condition where linear regression coefficients were applied.

5.2 LIMITATIONS

Climatology used for TCI

Ideally, TCI should be determined with a real climatology of the maximum LST. There are several limitations resulting from the characteristics of available data and the retrieval methodology.

The LST and SKT values around the pixel solar noon are extracted from different product timeslots. Given the temporal sampling of the variables (3-hourly in the case of SKT), the patterns of the respective spatial distribution clearly show longitudinal transition zones (Figure 4). Future improvements on the LST10 products will mitigate this limitation by using the more recent ERA5 Land reanalysis, with higher temporal and spatial resolution (9 km and 1 hour, respectively). The region now covered by the MSG IODC mission would benefit from a

reprocessing, since TCI values can only be derived from $SKT_{corrected}$, given no IODC data existed when the regression coefficients in Eq. (1) were derived.

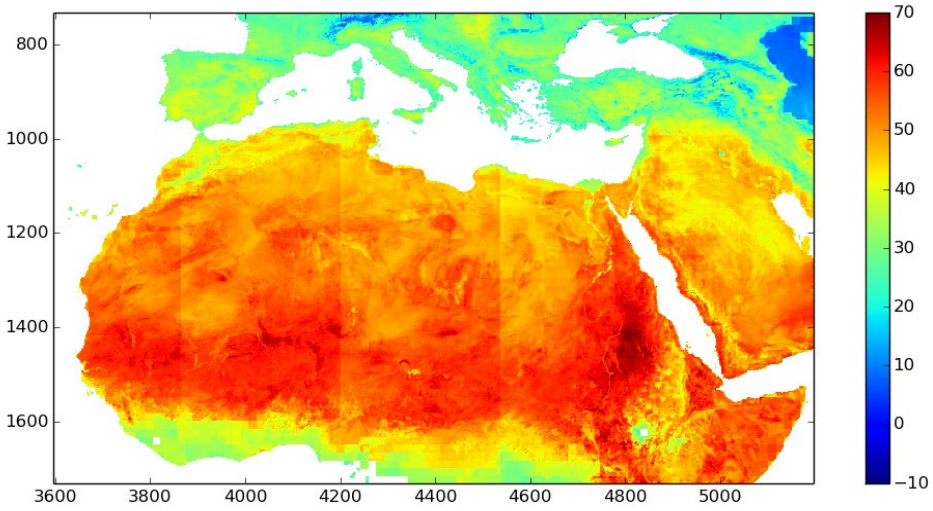


Figure 4- Example of the longitudinal discrepancies in LST climatological maxima (1985-2015 climatology for the dekad starting on March 21st).

The methodology based on the linear regression between LST and *corrected SKT* is limited by the quality of the fit, particularly over specific regions (e.g., tropics and sub-tropics). Main limitations are the sample size (Figure 5) and correlation between SKT and LST (Figure 6), for the coexisting period.

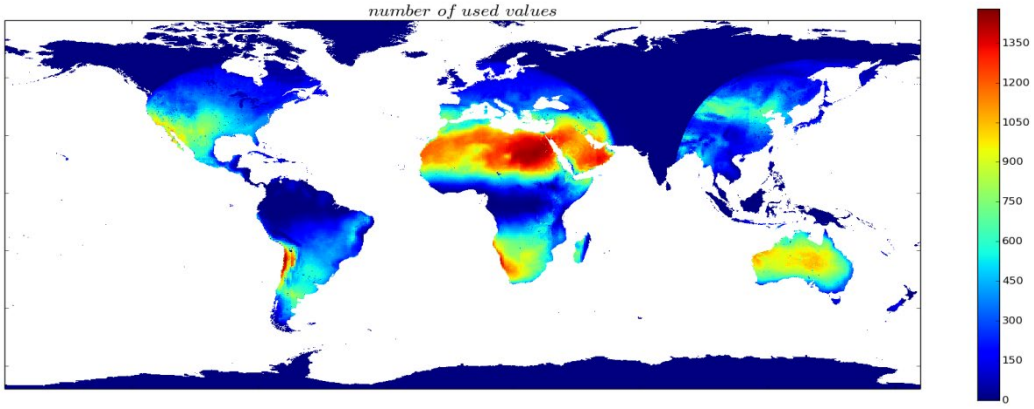


Figure 5: Sample size (N) used for linear regression between LST and *corrected SKT*, for the period 2010-2015.

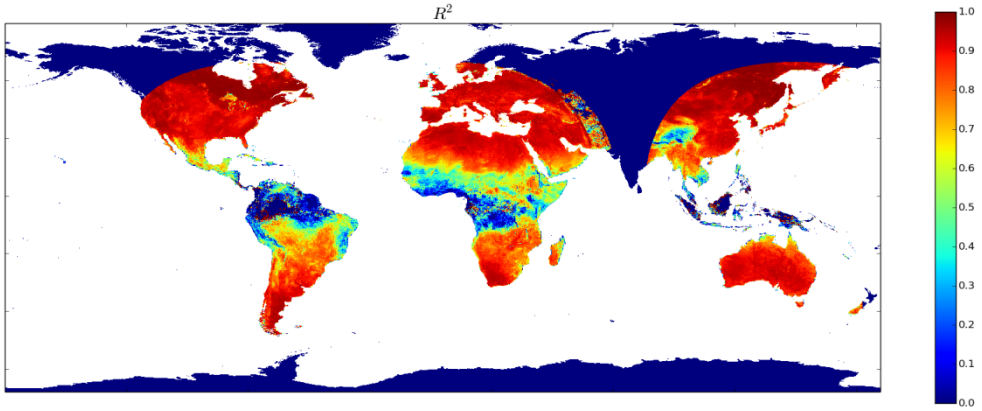


Figure 6: Coefficient of determination, R², for the period 2010-2015.

A number of recent studies have identified significant biases in reanalysis SKT, reaching values over 10 K over several regions during daytime (Johannsen et al., 2019; Nogueira et al., 2020; Orth et al., 2017; Trigo et al., 2015; Wang and Prigent, 2020). The differences in spatial and temporal resolution (e.g., 75 km and 3 h in the case of ERA-Interim versus 5 km and 1 h in the case of LST produced within CLMS), further enhance the discrepancies between satellite and reanalysis fields. Although these biases have been reduced in ERA5 and in ERA5 Land (Johannsen et al., 2019; Nogueira et al., 2020), the climatology used to derive CLMS TCI would greatly benefit from the use of these more up to date datasets. The linear regressions adjusted between satellite LST and re-analyses SKT are aimed at removing those biases.

Regardless of the mentioned limitations, geographic regions presenting low R² values are located at the ITCZ (Intertropical Convergence Zone) where cloud pixels are expected most of the time, hence TCI low values or TCI missing data, devaluating this issue.

Spatial Coverage

Since the LST product is only calculated under clear sky conditions it may happen that in some cloudy areas there are too few valid pixels to compute statistical parameters over the compositing period. However, the number of valid LST values available to calculate those statistics can be used as an indicator of the robustness of the LST10 product.

Figure 7 shows the percentage of valid LST values over one dekad in June 2019. The regions with less available data (corresponding to blue/green colours) correspond to cloudy regions and/or absence of satellite data. In those regions there is a high probability that LST10 product may not be available during at least part of the year.

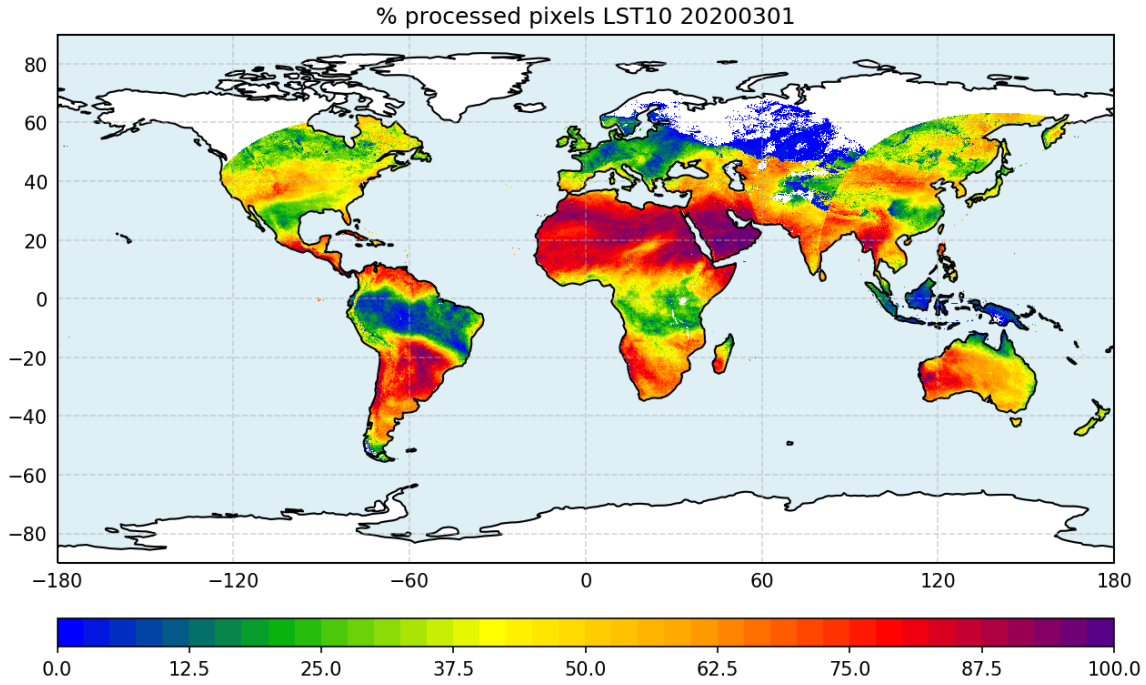


Figure 7: Percentage of valid LST pixels over the dekad starting on March 1st, 2020.

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6. RISK AND MITIGATION

The main risk associated to the LST10 product is the absence of its input, i.e., the CLMS LST product. As such, both products will benefit from the same mitigation procedures [CGLOPS1_ATBD_LST-V2.2].

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